

CBSE Test Paper 03
Chapter 6 Work Energy & Power

1. The potential energy of a spring whose spring constant is k and is compressed by x is **1**
 - a. $\frac{1}{4}kx^2$
 - b. $\frac{1}{2}kx^2$
 - c. kx^2
 - d. $2kx^2$
 2. Which of the following laws used to solve problems in collisions? **1**
 - a. Kirchhoff's laws
 - b. Newton's laws
 - c. kepler's laws
 - d. laws of conservation of linear momentum
 3. A bolt of mass 0.3 kg falls from the ceiling of an elevator moving down with an uniform speed of 7 m/s. It hits the floor of the elevator (length of the elevator = 3 m) and does not rebound. What is the heat produced by the impact? **1**
 - a. 9.22 J
 - b. 8.42 J
 - c. 8.82 J
 - d. 8.11 J
 4. For a ball dropped from a tower of height h the total mechanical energy is **1**
 - a. the difference of potential and kinetic energies
 - b. the potential energy
 - c. the sum of potential and kinetic energies
 - d. the kinetic energy
 5. Considering a completely inelastic collision of a particle of mass m_1 moving with initial velocity v_1 and a stationary particle of mass m_2 the final velocity is given by **1**
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- a. $\frac{m_2}{m_1+m_2} v_1$
- b. $\frac{m_1}{m_1+2m_2} v_1$
- c. $\frac{m_1}{m_1+m_2} v_1$
- d. $\frac{m_1}{2m_1+m_2} v_1$

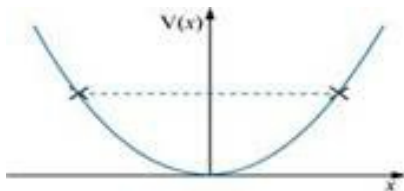
6. Is work done a scalar or a vector? **1**

7. If the momentum and total energy is conserved, which collision is occurred? **1**

8. In an elastic collision of two billiard balls, which of the following quantities remain conserved during the short time of collision of the balls (i.e., when they are in contact). **1**

- a. Kinetic energy.
- b. Total linear momentum? Give reason for your answer in each case

9. The potential energy function for a particle executing linear simple harmonic motion is given by $V(x) = \frac{kx^2}{2}$, where k is the force constant of the oscillator. For, $k = 0.5 \text{ Nm}^{-1}$, the graph of $V(x)$ versus x is shown in figure. Show that a particle of total energy 1 J moving under this potential must 'turn back' when it reaches $x = \pm 2 \text{ m}$. **2**



10. What are the conditions for work to be zero? **2**

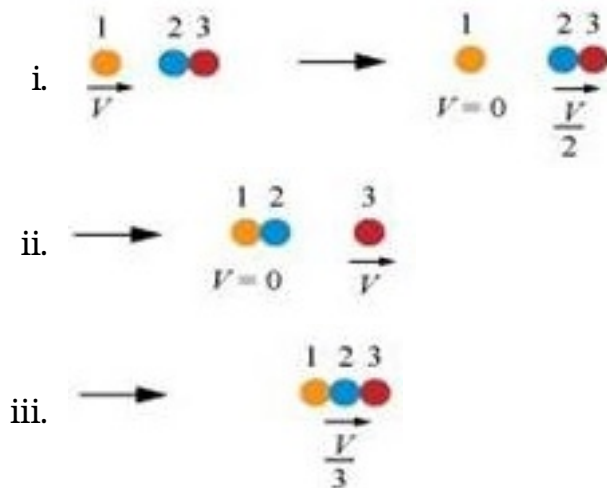
11. Calculate the work done by a car against gravity in moving along a straight horizontal road. The mass of the car is 400 kg and the distance moved is 2m. **2**

12. Consider a drop of small pebble of mass 1.00 g falling from a cliff of height 1.00 km. It hits the ground with a speed of 50.0 ms^{-1} . What is the work done by the unknown resistive force? **3**

13. An electron and a proton are detected in a cosmic ray experiment, the first with kinetic energy 10 keV, and the second with 100 keV. Which is faster, the electron or the proton? Obtain the ratio of their speeds. (electron mass = $9.11 \times 10^{-31} \text{ kg}$, proton

mass = 1.67×10^{-27} kg, 1 eV = 1.60×10^{-19} J). 3

14. Two identical ball bearings in contact with each other and resting on a frictionless table are hit head-on by another ball bearing of the same mass moving initially with a speed V . If the collision is elastic, which of the following figure is a possible result after collision? 3



15. A balloon filled with helium rises against gravity increasing its potential energy. The speed of the balloon also increases as it rises. How do you reconcile this with the law of conservation of mechanical energy? You can neglect the viscous drag of air and assume that the density of air is constant. 5